Lab Assignment 9

Object-Oriented Programming

Branch

B.E., 2^{nd} Year – ENC

THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

Submitted By:

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Batch: 2O23

1) Write a C++ program to write numbers 1 to 200 in a data file NUM TXT.

```
ques1.cpp ×
ques1.cpp > ...
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ofstream file("NUM.TXT");

    for(int i = 1; i <= 200; i++) {
        file << i << " ";
    }

    file.close();
    cout << "Numbers written successfully.";
    return 0;
}
```

Output:

```
PS C:\Users\Mridul\Desktop\OOPS> cd "c:\Users\Mridul\Desktop\OOPS\" ; if ($?) { g++ ques1.cpp
?) { .\ques1 }
Numbers written successfully.
PS C:\Users\Mridul\Desktop\OOPS> cd "c:\Users\Mridul\Desktop\OOPS\" ; if ($?) { g++ ques1.cpp
```

```
NUM.TXT
File   Edit   View   H1

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27
28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51
52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75
76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99
100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117
118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135
136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153
154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171
172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189
190 191 192 193 194 195 196 197 198 199 200

Ln 1, Col 1 | 692 characters | Plain text | 100% | Windows (CRLF | UTF-8
```

2) Write a user-defined function in C++ to read the content from a text file NOTES.TXT, count and display the number of alphabets present in it.

```cpp
#include <iostream>
#include <fstream>
#include <cctype>
using namespace std;

void countAlphabets() {
    ifstream file("NOTES.txt");

    if (!file) {
        cout << "File not found!";
        return;
    }

    char ch;
    int count = 0;

    while (file.get(ch)) {
        if (isalpha(ch)) count++;
    }

    cout << "Number of alphabets: " << count;
    file.close();
}

int main() {
    countAlphabets();
    return 0;
}
```

Output:

```
NOTES.txt
File    Edit    View

Mridulms
```

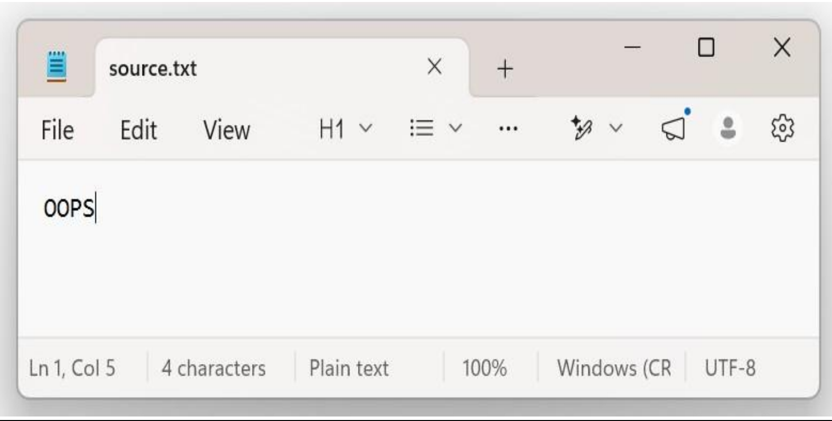
```
Number of alphabets: 8
--------------------------------
Process exited after 0.2903 seconds with return value 0
Press any key to continue . . .
```

3) Write a program to copy the contents of one file to another.

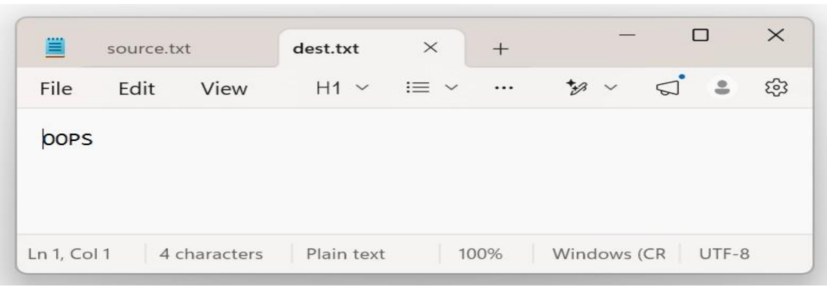
ques2.cpp ques3.cpp

```cpp
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    ifstream src("source.txt");
    ofstream dest("dest.txt");
    char ch;

    while (src.get(ch)) {
        dest.put(ch);
    }

    cout << "File copied successfully";
    src.close();
    dest.close();
    return 0;
}
```

Output:

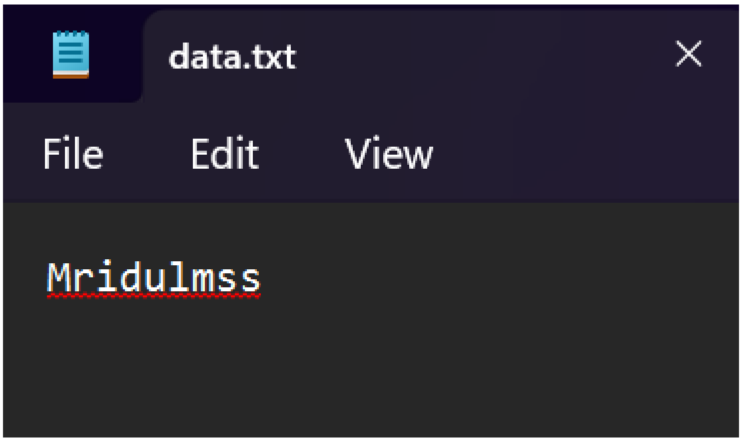



```
File copied successfully
--------------------------------
Process exited after 0.3019 seconds with return value 0
Press any key to continue . . .
```



4) Write a program to implement I/O operations on characters. I/O operations include inputting a string, calculating the length of the string, storing the string in a file and fetching the stored characters from it.

```cpp
#include <iostream>
#include <fstream>
#include <cstring>
using namespace std;

int main() {
    char str[100];

    cout << "Enter string: ";
    cin.getline(str, 100);

    cout << "Length: " << strlen(str) << endl;

    ofstream out("data.txt");
    out << str;
    out.close();

    ifstream in("data.txt");
    char ch;
    cout << "Stored string: ";

    while (in.get(ch)) {
        cout << ch;
    }

    in.close();
    return 0;
}
```



```
Enter string: Mridul
Length: 6
Stored string: Mridul
--------------------------------
Process exited after 4.793 seconds with return value 0
Press any key to continue . . .
```

5) Write a program to do the followings:

(a) Create a file A-Z and read 10th character using seekg().

(b) Overwrite 5th number using seekp().

(c) Find file size using tellg().

(d) Read last character using seekg().

(e) Create a file data.txt with multiple lines. Read the file and:

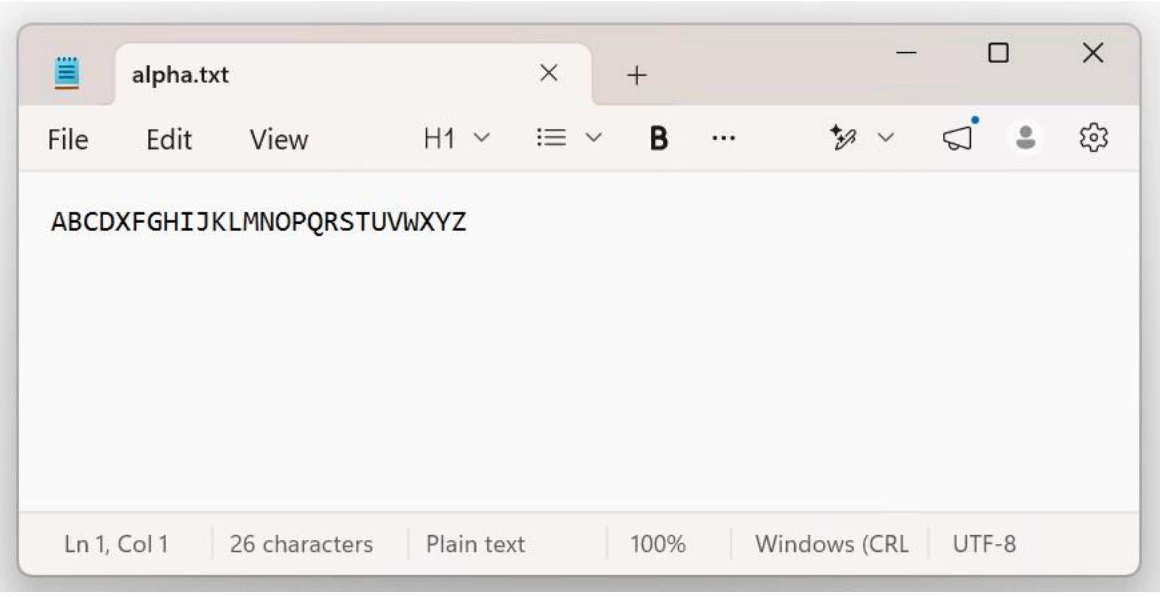
i. Move the pointer to the 10th byte using seekg()
ii. Display the current position using tellg().
iii. Read and display the remaining content.

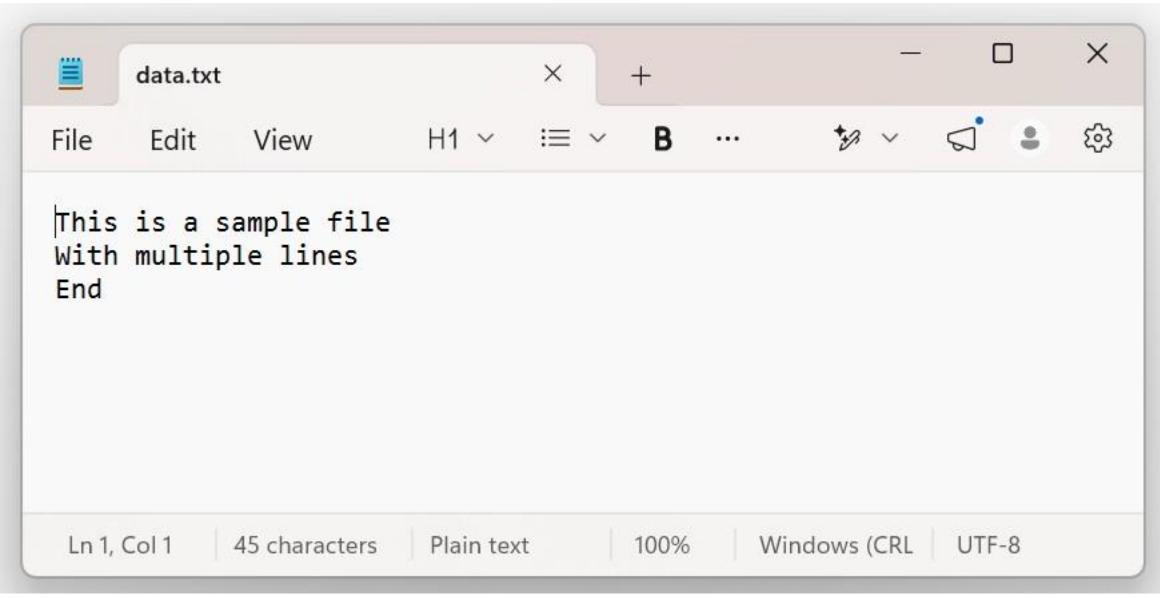
ques2.cpp ques3.cpp ques4.cpp ques5.cpp

```cpp
#include <iostream>
#include <fstream>
using namespace std;

int main() {
    // (a) Create file A-Z
    ofstream file("alpha.txt");
    for(char c='A'; c<='Z'; c++) file << c;
    file.close();

    ifstream in("alpha.txt");

    // (a) Read 10th character
    in.seekg(9);
    char ch;
    in.get(ch);
    cout << "10th char: " << ch << endl;

    in.close();

    // (b) Overwrite 5th character
    fstream f("alpha.txt", ios::in | ios::out);
    f.seekp(4);
    f.put('X');

    // (c) File size
    f.seekg(0, ios::end);
    cout << "File size: " << f.tellg() << endl;

    // (d) Last character
    f.seekg(-1, ios::end);
```

```cpp
    f.get(ch);
    cout << "Last char: " << ch << endl;

    f.close();

    // (e) data.txt operations
    ofstream d("data.txt");
    d << "This is a sample file\nWith multiple lines\nEnd";
    d.close();

    ifstream d_in("data.txt");

    d_in.seekg(10);
    cout << "Position: " << d_in.tellg() << endl;

    cout << "Remaining content:\n";
    while (d_in.get(ch)) {
        cout << ch;
    }

    d_in.close();
    return 0;
}
```

Output:





```
10th char: J
File size: 26
Last char: Z
Position: 10
Remaining content:
sample file
With multiple lines
End
--------------------------------
Process exited after 0.4852 seconds with return value 0
Press any key to continue . . .
```

6) Write a program to do the followings:

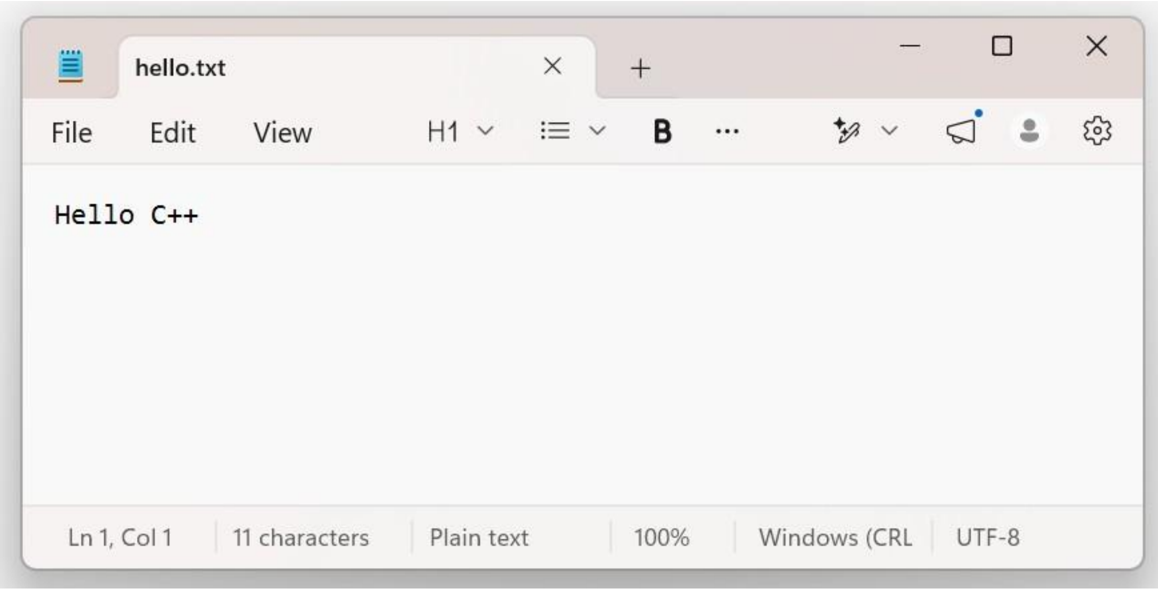
i. Create a file with text "Hello World" ii. After writing each character, display the current position of the put pointer using tellp().

iii. Replace "World" with "C++" using random access

ques2.cpp ques3.cpp ques4.cpp ques5.cpp ques6.cpp

```
#include <iostream>
#include <fstream>
using namespace std;
int main() {
    ofstream create("hello.txt");
    create.close();
    fstream file("hello.txt", ios::in | ios::out);
    string text = "Hello World";
    for (int i = 0; i < text.length(); i++) {
        file.put(text[i]);
        cout << "Position: " << file.tellp() << endl;
    }
    file.seekp(6);
    file << "C++  ";
    file.seekg(0);
    char ch;
    cout << "Final content: ";
    while (file.get(ch)) cout << ch;

    file.close();
    return 0;
}
```

<u>Output:</u>




```
Position: 1
Position: 2
Position: 3
Position: 4
Position: 5
Position: 6
Position: 7
Position: 8
Position: 9
Position: 10
Position: 11
Final content: Hello C++
--------------------------------
Process exited after 0.2975 seconds with return value 0
Press any key to continue . . .
```